No.	Status	Date
HIS-2011-0207-DS-4-RA-102	Issued	2011-02-18

UT Thickness Data Sheet - 4-RA-102

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 4-RA-102 located within the ATU-100. The primary objective is to document the current wall thickness and calculate the remaining life of the asset based on established site retirement criteria.

Kevin Doucette, acting as the lead UT technician (SNT-TC-1A UT Level II) for Halward Inspection Services Ltd., performed the measurements and data analysis. The technician is responsible for the accuracy of the readings and the application of the specified measurement procedures.

Technical Specifications

Line 4-RA-102 is constructed from A106 Gr.B with a nominal thickness of 8.56 mm and an outer diameter of 114.3 mm. The design limits for this rich amine service are 862 kPa(g) and 82 °C.

The minimum allowable thickness (t-min) for this line is 3.81 mm. Any reading recorded below this threshold is subject to an NCR per site rules.

Procedural Compliance

Thickness measurements were conducted in accordance with ASTM E797. The field activities were performed under the requirements of QSCP-NDT-001 Rev 2, which includes specific mandates for insulation removal, reinstatement, and inspection-port requirements as part of the CUI program.

The inspection was further governed by HIS-UT-002 Rev 1, the ultrasonic thickness measurement procedure submitted by Halward Inspection Services Ltd. for QSCP acceptance.

Inspection Results and Analysis

The data recorded in this sheet supports the findings detailed in HIS-2011-0207. This 5-year Class 1 survey identifies the current state of the rich amine line within the ATU-100.

Based on the calculated corrosion rate, the rich amine service on 4-RA-102 is currently at half remaining life. The next thickness measurement is scheduled according to the remaining life calculation and the Class 1 maximum interval.

Thickness readings, measured 2011-02-18

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2006-02-24	6.59	6.39	0.049	0.040	>50	2016-02-18	
6-SG-101-02	7.11	3.56	2006-02-24	6.51	6.14	0.061	0.074	34.7	2016-02-18	
6-SG-101-03	7.11	3.56	2006-02-24	6.42	6.17	0.061	0.050	42.9	2016-02-18	
6-SG-101-04	7.11	3.56	2006-02-24	6.28	6.05	0.070	0.046	35.8	2016-02-18	
6-SG-101-05	7.11	3.56	2006-02-24	6.22	5.87	0.085	0.070	27.2	2016-02-18	
6-SG-101-06	7.11	3.56	2006-02-24	6.49	6.27	0.054	0.044	>50	2016-02-18	
6-SG-101-08	7.11	3.56	—	—	6.33	—	—	—	2016-02-18	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2006-02-24	6.26	5.75	0.093	0.102	21.4	2016-02-18	
6-SG-102-02	7.11	3.56	2006-02-24	6.25	5.80	0.088	0.090	24.8	2016-02-18	
6-SG-102-03	7.11	3.56	2006-02-24	6.26	5.80	0.083	0.092	24.3	2016-02-18	
6-SG-102-04	7.11	3.56	2006-02-24	6.22	5.73	0.084	0.098	22.1	2016-02-18	
6-SG-102-05	7.11	3.56	2006-02-24	6.72	6.46	0.045	0.052	>50	2016-02-18	
6-SG-102-06	7.11	3.56	—	—	6.41	—	—	—	2016-02-18	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2006-02-24	8.27	8.32	0.013	-0.010	>50	2021-02-18	
4-RA-101-02	8.56	3.81	2006-02-24	8.27	8.20	0.023	0.014	>50	2021-02-18	
4-RA-101-03	8.56	3.81	2006-02-24	8.10	7.98	0.042	0.024	>50	2021-02-18	
4-RA-101-04	8.56	3.81	2006-02-24	8.34	8.21	0.028	0.026	>50	2021-02-18	
4-RA-101-05	8.56	3.81	2006-02-24	8.29	8.16	0.029	0.026	>50	2021-02-18	
4-RA-101-07	8.56	3.81	2006-02-24	8.30	8.30	0.020	0.000	>50	2021-02-18	
4-RA-101-08	8.56	3.81	2006-02-24	8.07	8.02	0.044	0.010	>50	2021-02-18	
4-RA-101-09	8.56	3.81	2006-02-24	8.43	8.32	0.017	0.022	>50	2021-02-18	
4-RA-101-10	8.56	3.81	2006-02-24	8.27	8.28	0.026	-0.002	>50	2021-02-18	
4-RA-101-11	8.56	3.81	2006-02-24	8.14	8.04	0.043	0.020	>50	2021-02-18	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2006-02-24	8.41	8.33	0.014	0.016	>50	2021-02-18	
4-RA-102-02	8.56	3.81	2006-02-24	8.15	7.99	0.039	0.032	>50	2021-02-18	
4-RA-102-03	8.56	3.81	2006-02-24	8.27	8.04	0.035	0.046	>50	2021-02-18	
4-RA-102-04	8.56	3.81	2006-02-24	8.24	7.96	0.039	0.056	>50	2021-02-18	
4-RA-102-05	8.56	3.81	2006-02-24	8.19	8.07	0.032	0.024	>50	2021-02-18	
4-RA-102-06	8.56	3.81	2006-02-24	8.15	8.01	0.033	0.028	>50	2021-02-18	
4-RA-102-07	8.56	3.81	2006-02-24	8.29	8.10	0.032	0.038	>50	2021-02-18	
4-RA-102-08	8.56	3.81	2006-02-24	8.05	7.88	0.041	0.034	>50	2021-02-18	
4-RA-102-09	8.56	3.81	2006-02-24	8.18	7.98	0.038	0.040	>50	2021-02-18	
4-RA-102-11	8.56	3.81	—	—	8.19	—	—	—	2021-02-18	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2006-02-24	10.76	10.47	0.035	0.058	>50	2021-02-18	
6-RA-103-02	10.97	4.57	2006-02-24	10.50	10.23	0.058	0.054	>50	2021-02-18	
6-RA-103-03	10.97	4.57	2006-02-24	10.67	10.42	0.038	0.050	>50	2021-02-18	
6-RA-103-04	10.97	4.57	2006-02-24	10.40	9.97	0.068	0.086	>50	2021-02-18	
6-RA-103-05	10.97	4.57	2006-02-24	10.34	10.10	0.057	0.048	>50	2021-02-18	
6-RA-103-06	10.97	4.57	2006-02-24	10.74	10.56	0.030	0.036	>50	2021-02-18	
6-RA-103-07	10.97	4.57	2006-02-24	10.66	10.55	0.029	0.022	>50	2021-02-18	
6-RA-103-08	10.97	4.57	2006-02-24	10.58	10.44	0.035	0.028	>50	2021-02-18	
6-RA-103-09	10.97	4.57	2006-02-24	10.67	10.56	0.030	0.022	>50	2021-02-18	
6-RA-103-10	10.97	4.57	2006-02-24	10.27	10.05	0.065	0.044	>50	2021-02-18	
6-RA-103-13	10.97	4.57	—	—	10.45	—	—	—	2021-02-18	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2006-02-24	8.16	7.95	0.037	0.042	>50	2021-02-18	
4-RA-104-02	8.56	3.81	2006-02-24	8.23	8.14	0.027	0.018	>50	2021-02-18	
4-RA-104-03	8.56	3.81	2010-09-13	8.59	8.56	0.069	0.069	>50	2021-02-18	
4-RA-104-04	8.56	3.81	2010-09-13	8.52	8.60	-0.185	-0.185	>50	2021-02-18	
4-RA-104-05	8.56	3.81	2010-09-13	8.43	8.57	-0.324	-0.324	>50	2021-02-18	
4-RA-104-06	8.56	3.81	2006-02-24	8.09	7.98	0.041	0.022	>50	2021-02-18	
4-RA-104-07	8.56	3.81	2006-02-24	7.97	7.78	0.054	0.038	>50	2021-02-18	
4-RA-104-08	8.56	3.81	—	—	8.18	—	—	—	2021-02-18	

Sign-off	Name	Function	Date	Signature
Prepared	Kevin Doucette	lead UT technician, SNT-TC-1A UT Level II	2011-02-18	_____